

Range of gof is image of (range of f under g)

- If $f : A \rightarrow B$ and $g : B \rightarrow C$ are two functions
 gof is defined for every case



But fog is defined iff

range of $g \leq \text{domain of } f$

fog maps from B to B i.e., $\text{fog} : B \rightarrow B$

$\text{fog} \neq \text{gof}$ (i.e., composition is not commutative)

$(\text{fog})\text{oh} = \text{fo}(\text{goh})$ (i.e., Associative)

$(\text{fog})^{-1} = g^{-1} \circ f^{-1}$ (Think why)

Let $f : A \rightarrow B$ be an invertible function then

$f^{-1} : B \rightarrow A$ is an inverse of f

$f^{-1} \circ f : A \rightarrow A$ is an identity function I_A

$f \circ f^{-1} : B \rightarrow B$ is an identity function I_B

Note:

- f is 1-1 & g is 1-1 $\Rightarrow \text{gof}$ is 1-1
- f is onto & g is onto $\Rightarrow \text{gof}$ is onto
- f is bijection & g is bijection $\Rightarrow \text{gof}$ is bijection
- If gof is onto then g is onto
- If gof is 1-1 then f is 1-1
- If gof is a bijection then f is onto $\Leftrightarrow g$ is 1-1

3.6.9 Partial Functions:

$f : A \rightarrow B$ is called partial function if ' f ' is defined only for some of $a \in A$.

The subset of A on which f is defined is called domain definition of f .

3.7 Groups:

3.7.1 Binary Operation:

The binary operator $*$ is said to be a binary operation on a non-empty set A if the set is closed under the operation.

i.e., $(a*b) \in A \quad \forall a, b \in A$

3.7.2 Binary Structure (or) Algebraic Structure:

A nonempty set A is called binary structure with respect to a binary operator $*$, if $*$ is binary operation on A . It is denoted as $(A, *)$

3.7.3 Semi Group:

$(A, *)$ is semigroup iff

- is closed operation
- is an associative property

3.7.4 Monoid:

$(A, *)$ is called monoid iff:

- (i) $*$ is closed operation
- (ii) $*$ is an associative property
- (iii) Identity element exists

3.7.5 Group:

$(A, *)$ is called monoid iff:

- (i) $*$ is closed operation
- (ii) $*$ is an associative property
- (iii) Identity element exists in A
- (iv) Every element of A has inverse

- Identity element if exists is unique.
- Inverse element if exist is unique.
- Finite Group:
A group with finite number of elements
- Order of a group: It is number of elements in the group.
- In a group of 2 element
 $a^{-1} = a, \forall a \in G$
- $(\{0, 1, 2, 3, \dots, m-1\}, \oplus_m)$ is a group
- Let S_n be set of positive integers less than n and relatively prime to n . then
 $(S_n, \otimes_n)$ is a group.
- Abelian Group:
A group $(G, *)$ is called abelian if $*$ is commutative.

3.7.6 Properties of Groups

- Let $(G, *)$ be a group
if $ab = ac \Rightarrow b = c$ (Left cancellation property)
 $ba = ca \Rightarrow b = c$ (Right cancellation property)
due to these property ever row and column in Caley table has exactly one element
- If G is a group and $a, b \in G$
then $(ab)^{-1} = b^{-1}a^{-1}$
- Group G is abelian $\Leftrightarrow (ab)^{-1} = b^{-1}a^{-1} \forall a, b \in G$
- $\forall a \in G, a^0 = e$

3.7.7 Homomorphism

- If $(G, *)$ and $(G', \oplus)$ are two groups then a function $f : G \rightarrow G'$ is called a homomorphism
if $f(a*b) = f(a) \oplus f(b)$
- If $f : G \rightarrow G'$ is a bijection, then we call the homomorphism isomorphism. It is denote as $G \cong G'$
- If e_1, e_2 are identity elements of G and G' respectively then $f(e_1) = e_2$
- If $a \in G$, and $f : G \rightarrow G'$ is a homomorphism $f(a^{-1}) = (f(a))^{-1}$

3.7.8 Order of an element:

The smallest positive integer n such that $a^n = e$ is called order of an element a in the group.

- The order of an element is divisor of order of the group.
- Order $(a) = \text{order}(a^{-1}) \quad \forall a \in G$
- $a^{-n} = b^n$ if $a^{-1} = b$ and n is any positive integer.

3.7.9 Subgroup:

A non-empty subset H of a group G is called subgroup of G if $(H, *)$ is also a group.

- For every group G , $\{e\}$ and G are called trivial subgroups of G .
- Every subgroup contains identity element of its parent group.
- If H is a subgroup of G then $\text{order}(H)$ divides $\text{order}(G)$ (Lagrange's theorem). The converse of Lagrange's theorem holds only for abelian groups.
- If H and k are two subgroups of same group, then
 $H \cap k$ is also, a subgroup
 $H \cup k$ need not to be a subgroup.
- If $H \subseteq G$, then H is called subgroup of G iff:
 (i) $(a*b) \in H \quad \forall a, b \in H$
 (ii) $a^{-1} \in H \quad \forall a \in H$
- If $H \subseteq G$, and if H is finite $\rightarrow$ (applies only when H is finite) and nonempty, then H is called subgroup iff:
 $(a*b) \in H \quad \forall a, b \in H$
- If G is a subgroup of composite order, then G necessarily has non-trivial subgroup.

3.7.10 Cyclic Groups:

A group G is called cyclic, if $\exists a \in G$ such that every element can be written as an integral power of a such an element 'a' is called generator.

- The order of generator is order of the group.
- If 'a' is a generator then a^{-1} is also a generator.
- All subgroups of a cyclic group are cyclic.
- If G is a cyclic group of order n , then no of generator of G is given by $\phi(n)$ (Euler's phi function)
- If 'a' is a generator of G , and let m be an integer such that $1 < m \leq n$ then

$$\text{order}(a^m) = \frac{n}{\gcd(n, m)}$$

- If G is a cyclic group with generator a and let d be divisor of $|G|$.
 For every d there exists exactly one subgroup of order d . this subgroup is generated by $a^{n/d}$ where $n = |G|$
 $\therefore$ No of subgroup of a cyclic group = no. of divisors of $|G|$ each subgroup is generated by $a^{n/d}$. (d is divisor of $|G|$)

3.7.11 Cyclic subgroup of a group:

If $(G, *)$ is any group and let $a \in G$.

The smallest subgroup of G containing a is $\langle a \rangle$ where

$$\langle a \rangle = \{a^n / \forall n \in \mathbb{Z}\}$$

Also $\langle a \rangle$ is a cyclic subgroup.

order of the subgroup $\langle a \rangle = \text{order of element 'a' in } G$.

- If H is any subgroup of G and if H contains 'a' then $\langle a \rangle \subseteq H$
- Thus if G is a group $\langle a \rangle$ is a cyclic subgroup of G , $\forall a \in G$.
Such subgroups are called generating sets

Note:

- Every group of prime order is cyclic in which every element (except e) is a generator element.
- Every cyclic group is abelian.
- A group is cyclic $\Leftrightarrow$ it can't be expressed as union of two proper subgroups.
- If $(G, *)$ and $(H, \oplus)$ are two groups
 $(G \times H, \bullet)$ is a group where $\bullet$ is defined as
 $(g_1, h_1) \bullet (g_2, h_2) = (g_1 * g_2, h_1 \oplus h_2)$
This group is called direct product of G and H .
- Every group of order less than or equal to 5 is abelian.
- There is only one unique group for each of the order 1,2,3.

